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TEACHING PHILOSOPHY

Visiting Instructor of Practice, Industrial Robotics, Utah Tech University

My teaching philosophy is centered on helping students move from learning individual technical skills to confidently applying those skills in complete engineering systems. My goal is for students to understand not only how to perform a task, but why it is done, how it connects to the larger system, and how they would troubleshoot or adapt that process in a professional environment. I want students to leave my courses with both technical knowledge and the confidence to work independently with unfamiliar equipment, documentation, and engineering problems.

DEVELOPING AND RESTRUCTURING THE CURRICULUM

When I began as Visiting Instructor of Practice at Utah Tech University, I helped develop a new Industrial Robotics curriculum and restructured the learning sequence so that students first encounter a concept in a practical setting, develop a deeper understanding of the underlying theory, and then apply that knowledge on physical industrial equipment. The restructured course teaches static and inertial forces to explain why robots behave as they do, then I/O and Ethernet communication so students can connect a robot to the rest of an automated system. This lets students see the relevance of coordinate systems, tool frames, programming, registers, I/O, PLC communication, and system commissioning. I place greater emphasis on technical documentation, troubleshooting, design iteration, and individual competency so that students learn to find and apply information instead of relying only on instructor-provided procedures.

I was also the first instructor to bring FANUC ROBOGUIDE into the course. ROBOGUIDE gives students offline programming and virtual workcell experience, so they can build, program, test, and troubleshoot a workcell before using the physical robots. That gives them more chances to practice and iterate, makes physical lab time more productive, and mirrors how industry commissions robotic cells. It reinforces a habit I care about: test and verify before you run.

HANDS-ON LEARNING WITH A PLC TRAINER

To teach how robots work inside a larger system, I helped develop a PLC trainer that pairs an AutomationDirect P1000 PLC with a FANUC LR Mate 200iD through CRMA58 terminal blocks. Students wire and configure group I/O, build digital logic, and establish Ethernet communication between the PLC and the robot controller. Working on real hardware makes the connection between a program, a signal, and a physical motion concrete in a way a diagram cannot.

SAFETY AS A CORE SKILL

I implemented lab safety protocols for student use of the industrial robotics equipment, and I treat them as a skill students practice rather than a list they sign. I model the behavior I expect: checking the workcell, confirming the robot's state, and keeping emergency stops in mind. In MTRN 2200 and MTRN 2205, which I am teaching for the first time, I focus on safe, practical industrial wiring and troubleshooting. My earlier work as a Makerspace intern, training users on CNC machines, laser engravers, and 3D printers and installing E-stops and circuit breakers, taught me that good training and good safety go together.

CONNECTING THE CLASSROOM TO INDUSTRY

My industry experience shapes what I teach. At Baxter Aerospace I developed an automated high-voltage test system for aerospace wire harnesses, integrated PLC control, programmed Yaskawa, FANUC, and Universal Robots, and built C# and SQL tools for MES work instructions. At IME Automation I contributed to an automated system project from development through completion. As the electrical and controls lead on my senior capstone with Lovesac, I designed the HMI, ladder logic, and panel wiring and worked through real process variability. I bring those examples of integration, troubleshooting, and documentation into the classroom.

GOALS GOING FORWARD

My broader goal is to continue developing courses that increasingly resemble the way engineering work is performed in industry. Students should have opportunities to design components, program and commission equipment, diagnose failures, document their work, and integrate multiple technologies into functioning systems. I am particularly interested in project-based experiences where students progress from structured introductory activities toward greater independence and system-level responsibility. Ultimately, I want my teaching to produce students who are curious, technically capable, willing to experiment, and prepared to approach complex engineering systems with a disciplined process of research, testing, troubleshooting, and continuous improvement.

CURRENT INSTRUCTIONAL DEVELOPMENT INITIATIVES

- **Robot I/O and PLC integration:** hands-on labs that progress from robot I/O and discrete wiring to Ethernet communication and PLC-coordinated robot operation
- **Complete robotic systems:** new lectures and labs using FANUC technical resources that emphasize end-of-arm tooling, robot setup, fault recovery, and system-level troubleshooting
- **FANUC ROBOGUIDE:** virtual workcell labs so students can build, program, test, and troubleshoot before using physical robots
- **MTRN 2200 and 2205:** evaluate the existing course structure, equipment, labs, and assessments, and use what I learn to improve future semesters
- **FANUC CERT Instructor Certification:** in progress through FANUC America